Algorithmic Trading and Quantitative Strategies

Part 6

Dr. Ayhan Yuksel, CFA, FRM, PRM

Trend Following Fundementals

What Is Trend Following

- A **trend** is the general direction of a market or of the price of an asset
- Trends can vary in length from short to intermediate, to long term.
- **Trend following**:
  - Trends have a tendency to continue, rather than reverse
  - "Go with the trend"

What Is Trend Following

Three Statistical Regimes

- Three regimes:
  - Trending
  - Mean reverting
  - Random
- Different strategies for different regimes:
  - Trending: ‘Buy High, Sell Higher’
  - Mean reverting: ‘Buy Low, Sell High’

Three Statistical Regimes

- Academic research has indicated that stock prices are on average very close to random walking. However, this does not mean that under certain special conditions, they cannot exhibit some degree of mean reversion or trending behavior.
- Furthermore, at any given time, stock prices can be both mean reverting and trending depending on the time horizon you are interested in.
- Constructing a trading strategy is essentially a matter of determining if the prices under certain conditions and for a certain time horizon will be mean reverting or trending, and what the initial reference price should be at any given time.

Trend Following Philosophy

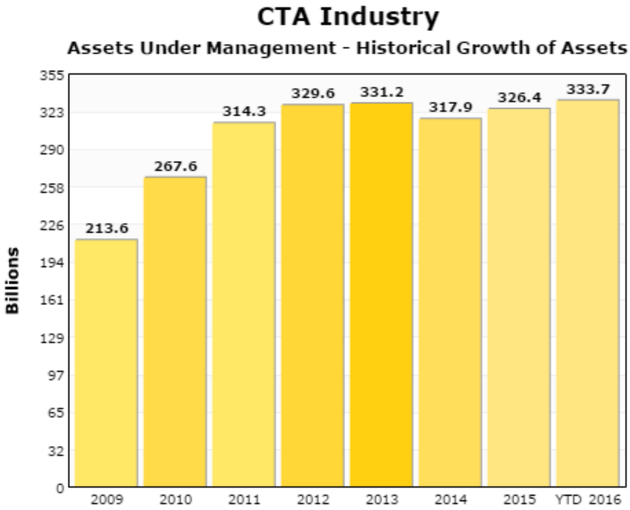
"Cut short your losses and let your profits run on" *(David Ricardo, 1772-1823)*

Trend Following Philosophy

- Wait for the emergence of a (bullish or bearish) trend
    - How do we know this?
- Take position along the direction of the trend
    - When and how much?
- Hold the position until the trend reverses
    - How do we know this?
    - Not 'Buy Low, Sell High', rather 'Buy High, Sell Higher'
    - Eat the middle beaf

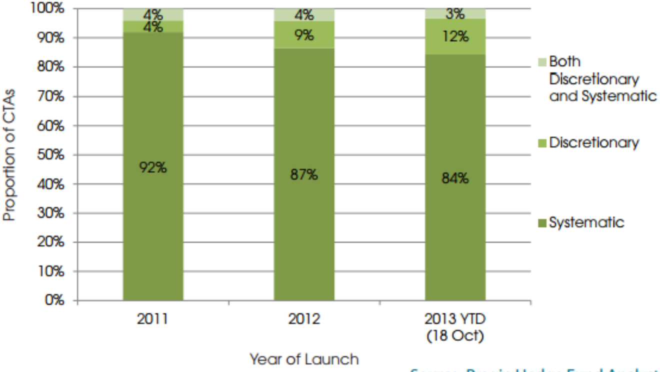
Popularity of Trend Following

CTA Industry - Assets Under Management



Popularity of Trend Following

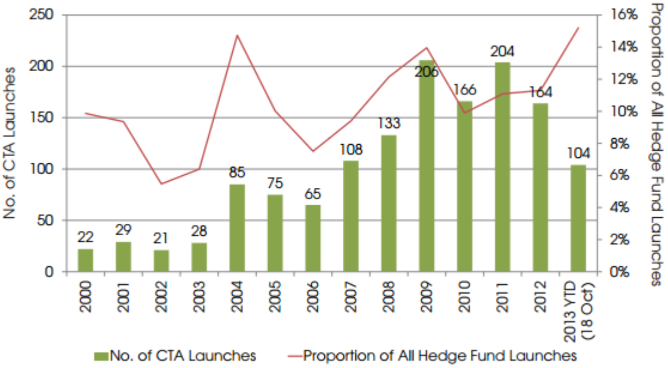
Fig. 2: Trading Methodology Employed by CTAs, 2011 - 2013 YTD (As at 18th October 2013)



Source: Preqin Hedge Fund Analyst

Popularity of Trend Following

Fig. 1: CTA Launches by Year of Inception and as a Proportion of All Fund Launches Each Year, 2000 - 2013 YTD (As at 18 October 2013)



Source: Preqin Hedge Fund Analyst

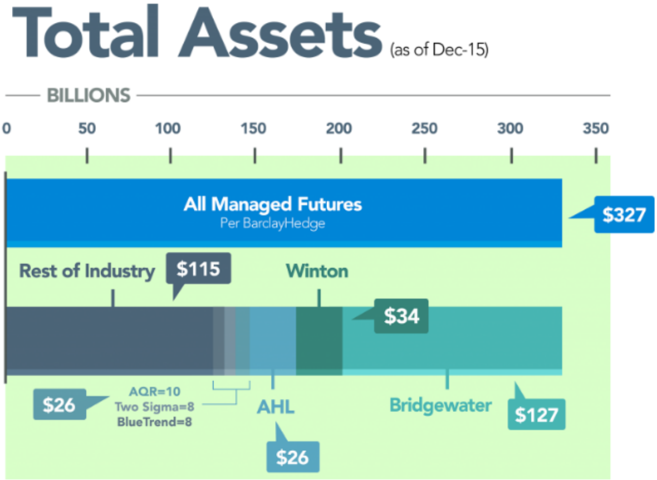
Popularity of Trend Following

Fig. 3: Top Five Known CTAs by Assets under Management (As at 18 October 2013)

Fund	Firm	Firm Location	Year of Inception	Fund AUM ($bn)	Strategy
AHL	Man Investments	UK	1996	12.5	Systematic
BlueTrend Master Fund	BlueCrest Capital	UK	2004	10.8	Systematic, Trend-Following
Winton Futures Fund Ltd	Winton Capital Management	UK	1997	9.7	Multi-Strategy
Diversified Trend Program	Transtrend	Netherlands	1995	7.1	Systematic, Trend-Following
Renaissance Institutional Diversified Alpha Fund	Renaissance Technologies	US	2012	5.0	US, Systematic

Source: Preqin Hedge Fund Analyst

Popularity of Trend Following



Does Trend Following Work?

A Century of Evidence on Trend-Following Investing

Executive Summary

We study the performance of trend-following investing across global markets since 1903, extending the existing evidence by more than 80 years. We find that trend-following has delivered strong positive returns and realized a low correlation to traditional asset classes each decade for more than a century. We analyze trend-following returns through various economic environments and highlight the diversification benefits the strategy has historically provided in equity bear markets. Finally, we evaluate the recent environment for the strategy in the context of these long-term results.

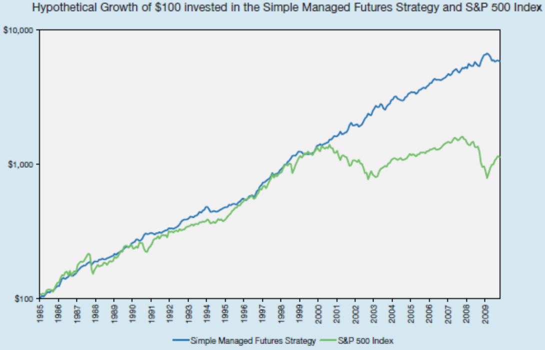
(Hurst, Ooi, and Pedersen 2012, 2010)

Does Trend Following Work?

strategy that is simple, without many of the often arbitrary choices of more complex models. Specifically, we construct an equal-weighted combination of 1-month, 3-month, and 12-month time series momentum strategies for 59 markets across 4 major asset classes – 24 commodities, 11 equity indices, 15 bond markets, and 9 currency pairs – from January 1903 to June 2012. Since not all markets have return data going back to 1903, we construct the strategies using the largest number of assets for which return data exist at each point in time. We use futures returns when they are available. Prior to the availability of futures data, we rely on cash index returns financed at local short rates for each country. Appendix A lists the markets that we consider and the source and length of historical return data used.

Does Trend Following Work?

Exhibit 4: The Performance of the Simple Managed Futures Strategy over time (gross of transaction costs).



Please read important disclosures relating to hypothetical performance at the end of this paper.

Does Trend Following Work?

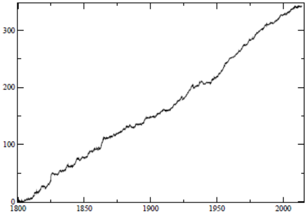


Figure 4: Aggregate performance of the trend on all sectors. t-stat=10.5, de-biased t-stat=9.8, Sharpe ratio=0.72.

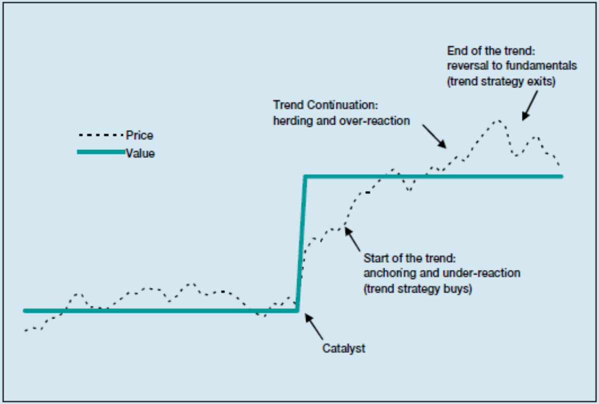
sector	SR (T)	t-stat (T)	t-stat (T*)	SR (μ)	t-stat (μ)	start date
Currencies	0.47	2.9	2.9	0.1	0.63	1973
Commodities	0.28	4.1	3.1	0.3	4.5	1800
Bonds	0.4	3.9	2.7	-0.1	-1	1918
Indices	0.7	10.2	6.3	0.4	5.7	1800

Table 8: Sharpe ratio and t-stat of the trend (T), of the de-biased trend (T*) and of the drift component μ of the different sectors. We also list the starting date for each sector

(Lempérière et al. 2014)

Why Trend Following Works

Exhibit 2: Stylized plot of the lifecycle of a trend.



Source: AQR. For illustrative purposes only.

Why Trend Following Works

Start of the Trend: Under-reaction to Information

- In the previous exhibit , a catalyst - e.g., a positive earnings release, a supply shock, or a demand shift - causes the value of an equity, commodity, currency, or bond to change. (The change in value is immediate, as shown by the solid blue line.) The market price (shown by the dotted black line) moves up as a result of the catalyst, but it initially under-reacts and therefore continues to go up for a while.

Why Trend Following Works

Start of the Trend: Under-reaction to Information

- Anchor-and-insufficient-adjustment: Edwards (1968), Tversky and Kahneman (1974) find that people anchor their views to historical data and adjust their views insufficiently to new information.
- The disposition effect: Shefrin and Statman (1985), Frazzini (2006) observe that people tend to sell winners too early and ride losers too long. They sell winners too early because they like to realize their gains. This selling creates downward price pressure, which slows down the upward price adjustment to the new fundamental level. On the other hand, people hang on to losers for too long since realizing losses is painful. Instead, they try to "make back" what has been lost. In this case, the absence of willing sellers keeps prices from adjusting downward as fast as they should.
- Non-profit-seeking market participants who fight trends: Silber (1994) argues that central banks operate in the currency and fixed-income markets to reduce exchange-rate volatility and manage inflation expectations, thus potentially slowing down the price-adjustment to news. As another example, hedging activity in commodity markets can also slow down price discovery.

Why Trend Following Works

Trend Continuation: Over-reaction

- Herding and feedback trading: De Long et al. (1990) and Bikhchandani et al. (1992) argue that when prices have moved up or down for a while, some traders may jump on the bandwagon, and this herding effect can feed on itself. Herding has been documented among equity analysts in their recommendations and earnings forecasts, in institutional investors' investment decisions, and in mutual fund investors who tend to move from funds with recent poor performance and herd into funds that have recently done well.
- Confirmation bias and representativeness: Wason (1960) and Tversky and Kahneman (1974) show that people tend to look for information that confirms what they already believe and look at recent price moves as representative of the future. This can lead investors to move capital into investments that have recently made money, and conversely out of investments that have declined, causing trends to continue.
- Risk management: Garleanu and Pedersen (2007) argue that some risk management schemes imply selling in down markets and buying in up markets, in line with the trend. For instance, stop-losses, or downsizing positions in a bear market as volatility (or Value at Risk) increases.

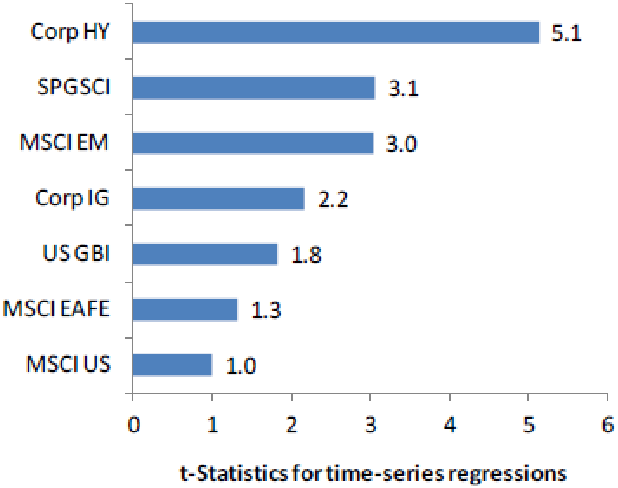
Why Trend Following Works

End of the Trend

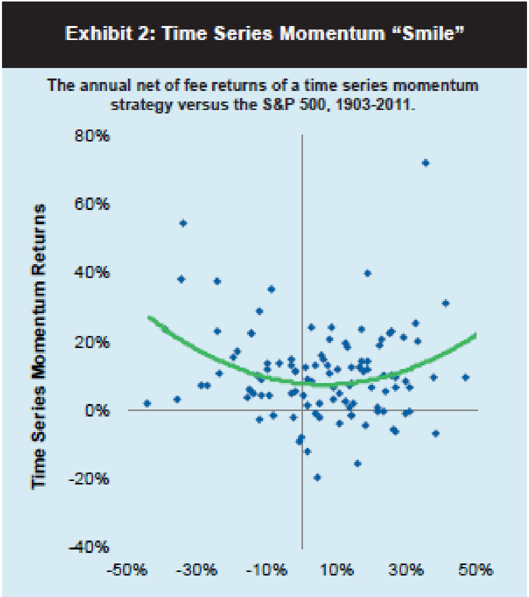
Obviously, trends cannot go on forever. At some point, prices extend beyond underlying fundamental value. As people start to realize that prices have gone too far, they revert towards fundamental value and the trend dies out. The market may become range bound until new events cause price moves and set off new trends. One of the main challenges for managed futures strategies is to minimize losses associated with the ending of trends and to preserve capital in range bound markets that do not exhibit trends.

Why Trend Following Works

Figure 31: Previous months' returns predicts next month's returns across Asset Classes (1989-2014)



Trend Following Smile



Diversification

Exhibit 1: Hypothetical Performance of Time Series Momentum

Strategy performance after simulated transaction costs both gross and net of hypothetical 2-and-20 fees.

Time Period	Gross of Fee Returns (Annualized)	Net of 2/20 Fee Returns (Annualized)	Realized Volatility (Annualized)	Sharpe Ratio, Net of Fees	Correlation to S&P 500 Returns	Correlation to US 10-year Bond Returns
Full Sample:						
Jan 1903 - June 2012	20.0%	14.3%	9.9%	1.00	-0.05	-0.05
By Decade:						
Jan 1903 - Dec 1912	18.8%	13.4%	10.1%	0.84	-0.30	-0.59
Jan 1913 - Dec 1922	17.1%	11.9%	10.4%	0.70	-0.12	-0.11
Jan 1923 - Dec 1932	17.1%	11.9%	9.7%	0.92	-0.07	0.10
Jan 1933 - Dec 1942	9.7%	6.0%	9.2%	0.66	0.00	0.55
Jan 1943 - Dec 1952	19.4%	13.7%	11.7%	1.08	0.21	0.22
Jan 1953 - Dec 1962	24.8%	18.4%	10.0%	1.51	0.21	-0.18
Jan 1963 - Dec 1972	26.9%	19.6%	9.2%	1.42	-0.14	-0.35
Jan 1973 - Dec 1982	40.3%	30.3%	9.2%	1.89	-0.19	-0.40
Jan 1983 - Dec 1992	17.8%	12.5%	9.4%	0.53	0.15	0.13
Jan 1993 - Dec 2002	19.3%	13.6%	8.4%	1.04	-0.21	0.32
Jan 2003 - June 2012	11.4%	7.5%	9.7%	0.61	-0.22	0.20

Source: AQR. Please read important disclosures at the end relating to hypothetical performance and risks.

Diversification

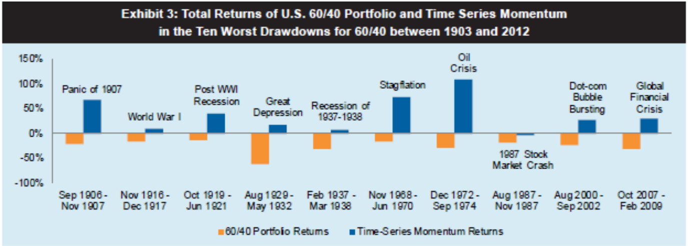
Exhibit 4: Diversifying 60/40 with an Allocation to Time Series Momentum

Performance characteristics of the 60/40 portfolio and a portfolio with 80% invested in the 60/40 portfolio and 20% invested in the time series momentum strategy, from January 1903 to June 2012.

	Total Net of Fee Returns (Annualized)	Realized Volatility (Annualized)	Sharpe Ratio, Net of Fees	Maximum Drawdown
60/40 Portfolio	8.0%	11.1%	0.34	-62%
80% 60/40 Portfolio, 20% Time Series Momentum Strategy	9.5%	9.0%	0.57	-52%

Source: AQR. Time Series performance is hypothetical as described above.

Diversification



Source: AQR. Time Series performance is hypothetical as described above.

Diversification

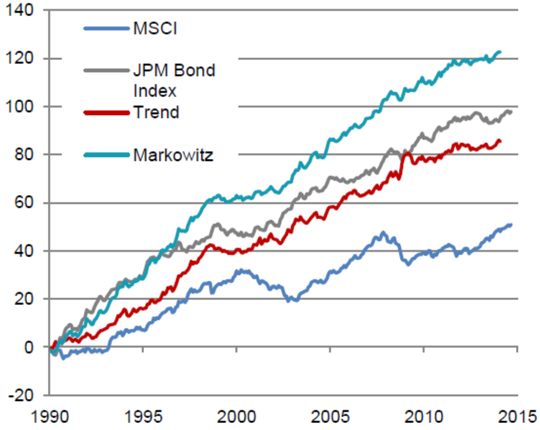


Figure 5: The result of "optimally" combining our trend following strategy with the J. P. Morgan bond index and the MSCI world equity index

Popular Trend Following Models

Moving Average Systems

MA Crossover

- $MA_t(n) = \sum_{i=0}^{n-1} P_{t-i}$
- $Ind_t = (MA_t(m) > MA_t(n))$
- $1 \leq m < n$
- Rules:
  - Long if $Cross(Ind_t, 1)$
  - Short if $Cross(0, Ind_t)$

Moving Average Systems

MA Crossover

Moving Average Systems

Double MA Crossover

- $Ind_t^L = (MA_t(m) > MA_t(n) > MA_t(p))$
- $Ind_t^S = (MA_t(m) < MA_t(n) < MA_t(p))$
- $1 \leq m < n < p$
- Rules:
  - Long if $Cross(Ind_t^L, 1)$
  - Short if $Cross(Ind_t^S, 1)$

Moving Average Systems

Double MA Crossover

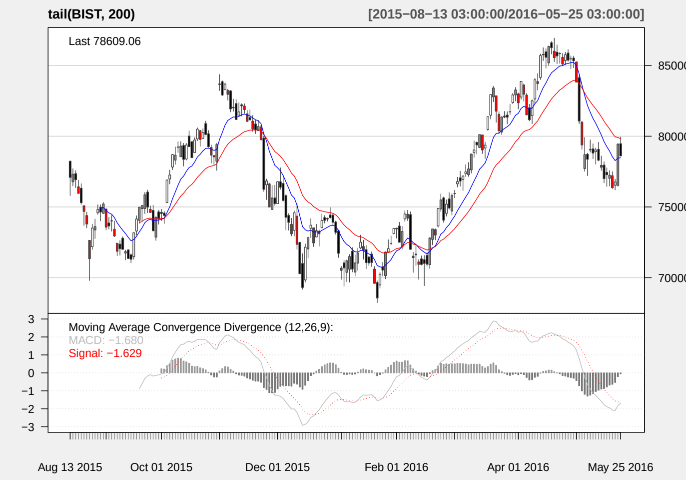
Moving Average Systems

MACD (Moving Average Convergence Divergence)

- MACD Line = EMA(Price,m)-EMA(Price,n)
- Signal Line = EMA(MACD Line,h)
- MACD Histogram = MACD Line - Signal Line
- Typical values: m=12, n=26, h=9
- Rules:
  - Signal Crossover
    - Long if Cross(MACD Line, Signal Line)
    - Short if Cross(Signal Line, MACD Line)
  - Centerline Crossovers
    - Long if Cross(MACD Line,0)
    - Short if Cross(0,MACD Line)

Moving Average Systems

MACD



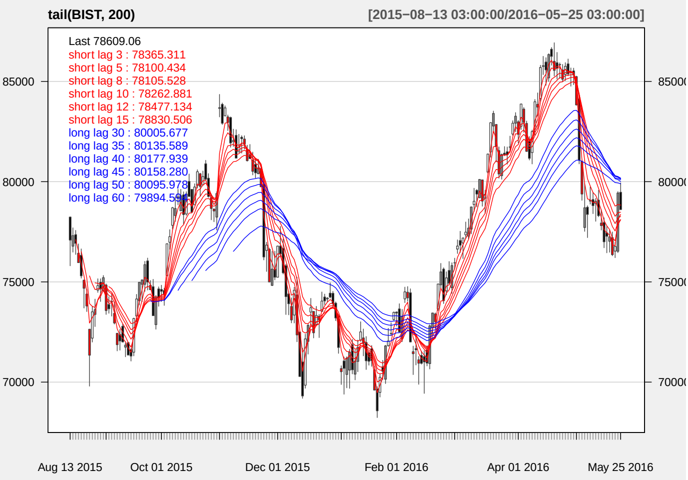
Moving Average Systems

Trend Strength Indicator I

- Draw large number of moving averages with varying lag lengths: $MA(n), n \in N = \{n_1, n_2, \ldots, n_k\}$
- Count the number of MA's below or above the price
- $Ind = \frac{1}{k}\sum_{i=1}^{k}(P > MA(n_i)) - (P < MA(n_i))$
- Rules:
  - Long if Cross(Ind , Long Threshold)
  - Short if Cross(Short Threshold,Ind)

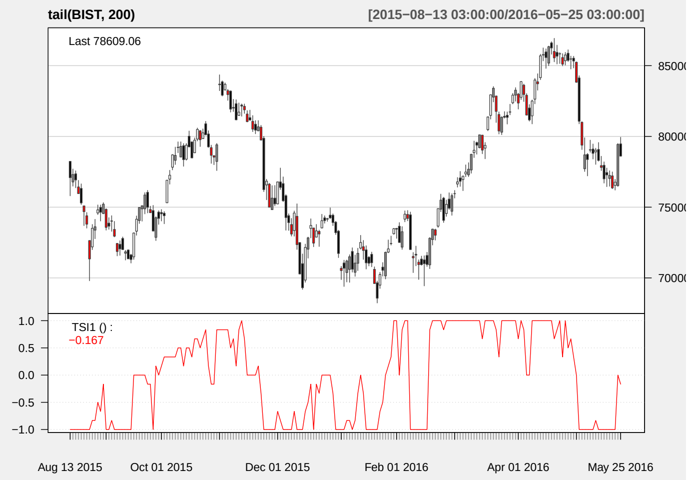
Moving Average Systems

Trend Strength Indicator I



Moving Average Systems

Trend Strength Indicator I



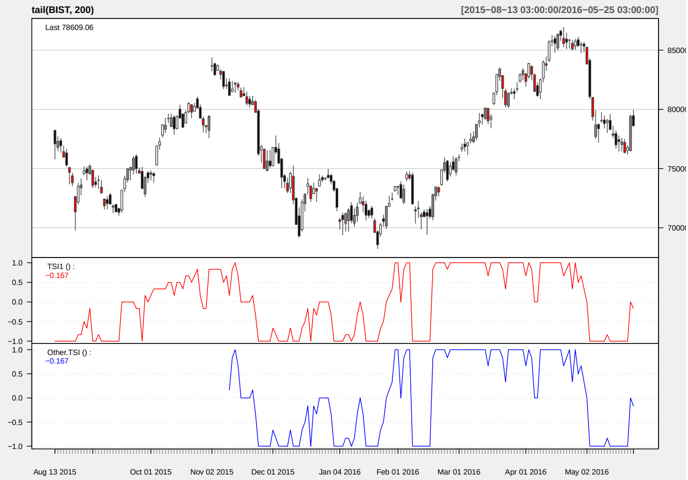
Moving Average Systems

Trend Strength Indicator II

- Draw large number of moving averages with varying lag lengths: $MA(n), n \in N = \{n_1, n_2, \ldots, n_k\}$
- Count the number of increasing/decreasing MA's
- $Ind = \frac{1}{k}\sum_{i=1}^{k}(MA_t(n_i) > MA_{t-1}(n_i)) - (MA_t(n_i) < MA_{t-1}(n_i))$
- Rules:
  - Long if Cross(Ind , Long Threshold)
  - Short if Cross(Short Threshold,Ind)

Moving Average Systems

Trend Strength Indicator II



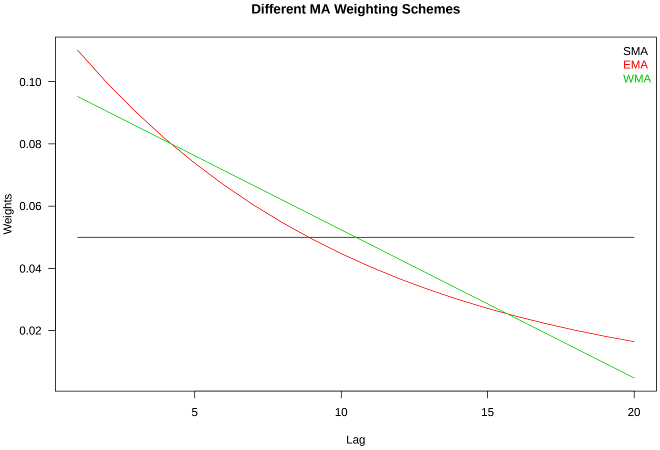
Moving Average Systems

Averaging Techniques

- Simple:
  - $SMA_t = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}$
- Exponential:
  - $EMA_t = \lambda P_t + (1-\lambda)EMA_{t-1}$
  - $\lambda = \frac{2}{1+n}$
- Weighted:
  - $WMA_t = \sum_{i=0}^{n-1} \frac{n-i}{n(n+1)/2} P_{t-i}$

Moving Average Systems

Averaging Techniques



Moving Average Systems

Averaging Techniques

Time Series Momentum Systems

Past Return

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Time series momentum ☆

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A B S T R A C T

We document significant "time series momentum" in equity index, currency, commodity, and bond futures for *each* of the 58 liquid instruments we consider. We find persistence in returns for one to 12 months that partially reverses over longer horizons, consistent with sentiment theories of initial under-reaction and delayed over-reaction. A diversified portfolio of time series momentum strategies across all asset classes delivers substantial abnormal returns with little exposure to standard asset pricing factors and performs best during extreme markets. Examining the trading activities of speculators and hedgers, we find that speculators profit from time series momentum at the expense of hedgers.

(Moskowitz, Ooi, and Pedersen 2012)

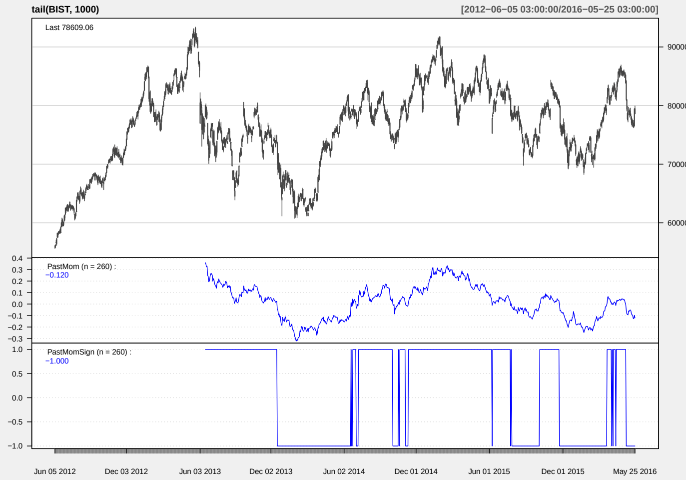
Time Series Momentum Systems

Past Return

- $r_t = log(\frac{P_t}{P_{t-1}})$
- $r_{t-h:t} = \sum_{i=0}^{h-1} r_{t-i}$
- $Ind_t = sign(r_{t-h:t})$
- Rules:
  - Long if $Cross(Ind_t, 1)$
  - Short if $Cross(-1, Ind_t)$

Time Series Momentum Systems

Past Return



Time Series Momentum Systems

Past Risk-Adjusted Return

Risk-Adjusted Time Series Momentum*

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September 25, 2014

Abstract

We introduce a new class of momentum strategies, the risk-adjusted time series momentum (RAMOM) strategies, which are based on averages of past futures returns, normalized by their volatility. We test these strategies on a universe of 64 liquid futures contracts and show that RAMOM strategies outperform the time series momentum (TSMOM) strategies of Ooi, Moskowitz, and Pedersen (2012) for almost all combinations of holding and look-back periods. This outperformance is driven by the following new striking stylized fact that we document: For almost all of the 64 futures contracts, independent of the asset class, realized futures volatility is contemporaneously negatively related to the Fama and French (1987) market (MKT), value (HML), and momentum (UMD) factors. As a result, RAMOM returns have a natural, built-in exposure to the MKT, HML, and UMD factors and

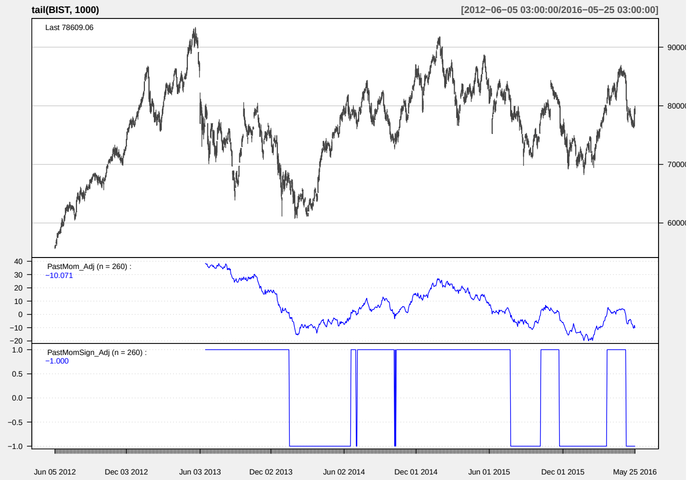
Time Series Momentum Systems

Past Risk-Adjusted Return

- $r_t = log(\frac{P_t}{P_{t-1}})$
- $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda) r_t^2$
- $\tilde{r}_t = r_t / \sigma_t$
- $\tilde{r}_{t-h:t} = \sum_{i=0}^{h-1} \tilde{r}_{t-i}$
- $Ind_t = sign(\tilde{r}_{t-h:t})$
- Rules:
  - Long if $Cross(Ind_t, 1)$
  - Short if $Cross(-1, Ind_t)$

Time Series Momentum Systems

Past Risk-Adjusted Return



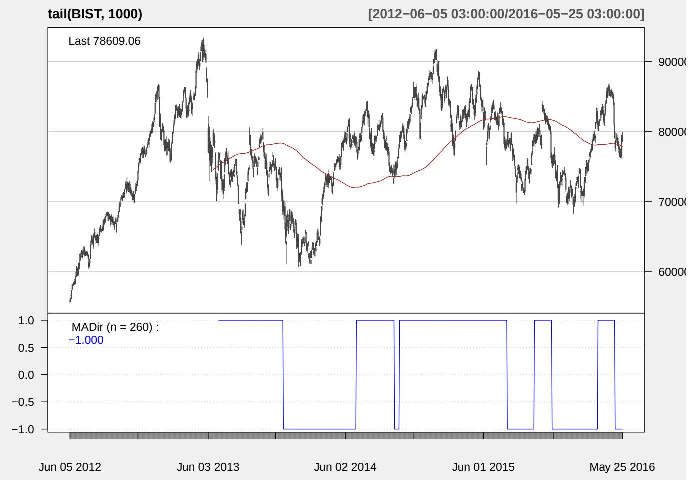
Time Series Momentum Systems

Direction of MA

- Calculate MA on price: $MA_t(n)$
- Calculate the direction of MA:
  - $Ind = sign(MA_t(n) - MA_{t-1}(n))$
  - $Ind = sign(MA_t(n) - MA_{t-h}(n))$
  - Other methods
- Rules:
  - Long if $Cross(Ind_t, 1)$
  - Short if $Cross(-1, Ind_t)$

Time Series Momentum Systems

Direction of MA



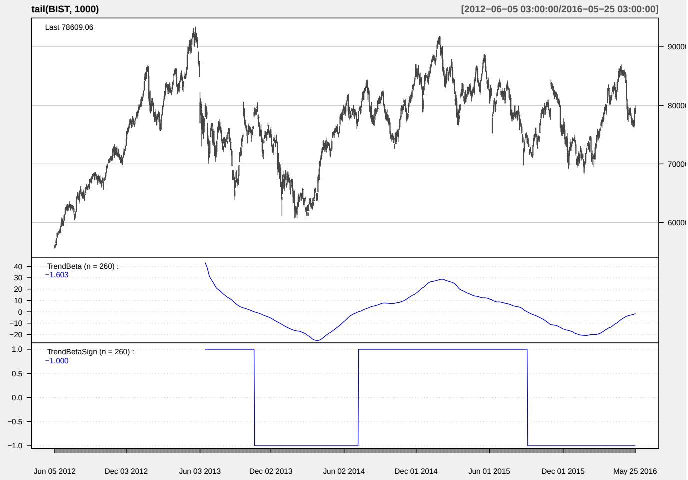
Time Series Momentum Systems

Linear Trend

- Fit a time trend to recent prices:
  - $P_t = \alpha + \beta t + \epsilon_t$
- Calculate the t-statistics for β
  - $Ind = tstat(\beta)$
- In order to account for potential autocorrelation and heteroskedasticity in the price process, Newey and West t-statistics are used.
- Rules:
  - Long if $Cross(Ind, 2)$
  - Short if $Cross(-2, Ind)$

Time Series Momentum Systems

Linear Trend



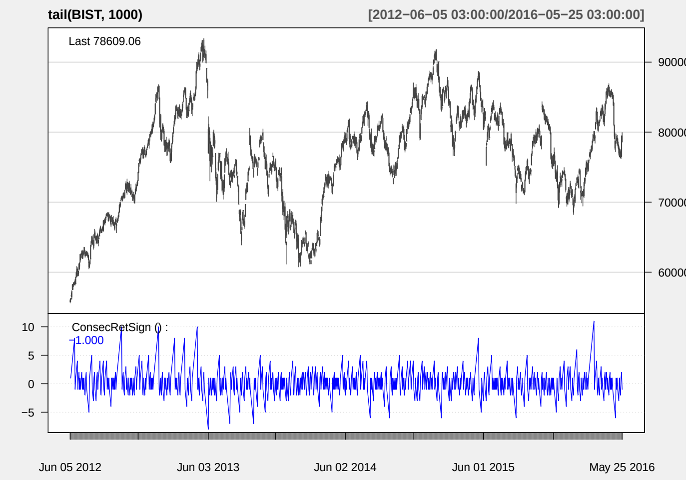
Time Series Momentum Systems

Consecutive Return Sign

- Calculate the number of consecutive positive/negative returns:
  - $N_{Positive}$
  - $N_{Negative}$
- Frequency may be daily/weekly/monthly
- Rules:
  - Long if $Cross(N_{Positive}, LongThreshold)$
  - Short if $Cross(N_{Short}, ShortThreshold)$

Time Series Momentum Systems

Consecutive Return Sign



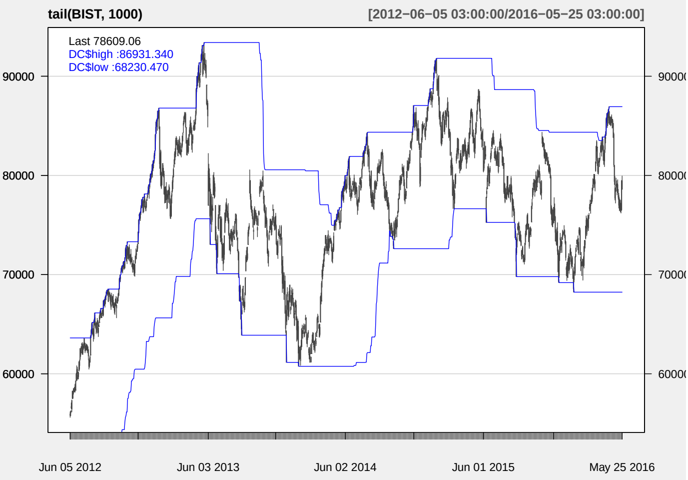
Breakout Systems

Donchian Channel Breakout

- Breakout trading:
  - Define a trading range:
    - Donchian channel
    - Bollinger bands
    - Support & resistance levels
  - Wait for a breakout
  - Breakout indicates starting of a new trend
  - Take position along the direction of the trend
- Donchian Channel Breakout
  - Upper Band: $Up_t(n) = max(H_{t-n}, \cdots, H_{t-1})$
  - Lower Band: $Low_t(n) = min(L_{t-n}, \cdots, L_{t-1})$
  - Rules:
    - Long if $Cross(P_t, Up_t(n))$
    - Short if $Cross(Low_t(n), P_t)$

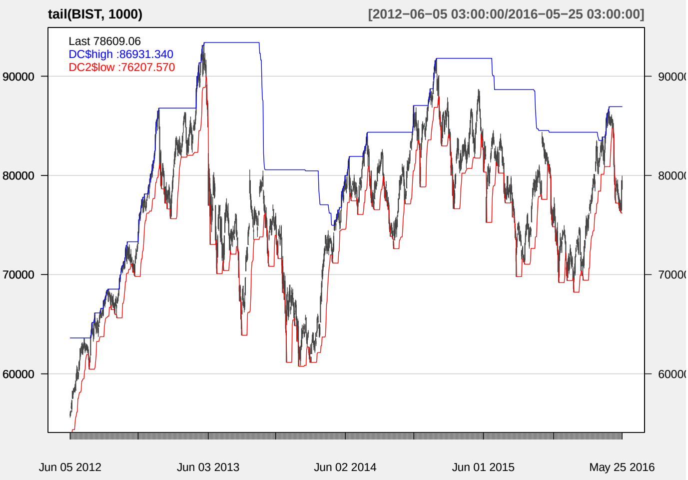
Breakout Systems

Donchian Channel Breakout



Breakout Systems

Donchian Channel Breakout



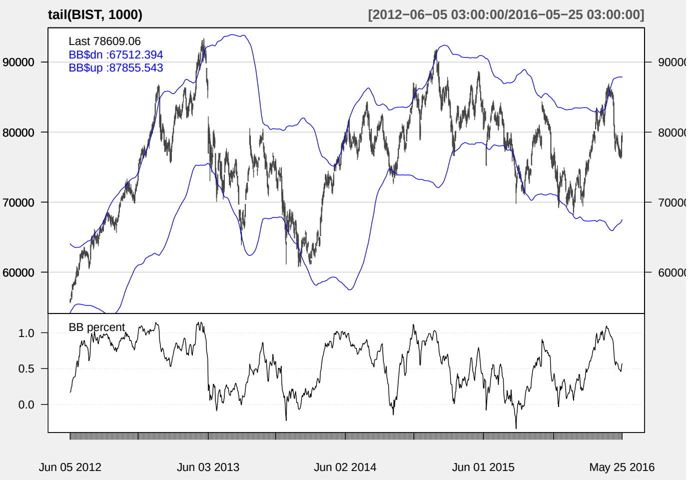
Breakout Systems

Bollinger Band Breakout

- Calculate average price: $MA_t(n) = \sum_{i=0}^{n-1} P_{t-i}$
- Calculate price volatility: $V_t(n) = StDev(P_{t-n+1}, \cdots, P_t)$
- Upper Band: $Up_t(n) = MA_t(n) + s \cdot V_t(n)$
- Lower Band: $Low_t(n) = MA_t(n) - s \cdot V_t(n)$
- Rules:
  - Long if $Cross(P_t, Up_t(n))$
  - Short if $Cross(Low_t(n), P_t)$

Breakout Systems

Bollinger Band Breakout



Breakout Systems

Turtle System

http://www.investopedia.com/articles/trading/08/turtle-trading.asp

Selected Issues on Trend Following

Equivelance of Trend Following Systems

Which Trend is Your Friend?

Ari Levine and Lasse Heje Pedersen*

Abstract

Managed-futures funds and CTAs trade predominantly on trends. There are several ways of identifying trends, either using heuristics or statistical measures often called "filters." Two important statistical measures of price trends are time series momentum and moving average crossovers. We show both empirically and theoretically that these trend indicators are closely connected. In fact, they are equivalent representations in their most general forms, and they also capture many other types of filters such as the HP filter, the Kalman filter, and all other linear filters. Further, we show how these filters can be represented through "trend signature plots" showing their dependence on past prices and returns by horizon. Our results unify and broaden a range of trend-following strategies and we discuss the implications for investors.

(Levine and Pedersen 2016)

Equivelance of Trend Following Systems

A TSMOM strategy goes long when prices have been moving up, and short when prices have been moving down. The simplest TSMOM signal is the past return over some time period, say m months or days:

$$TSMOM_t^m = return_{t-m,t} \tag{1}$$

For instance, 12-month momentum considers the return over the past 12 months. The return can be computed as the ratio of prices, P_t / P_{t-m}, or of a return index that takes dividends or coupons into account for cash instruments, and handles roll yields and implicit financing for futures. Alternatively, it can be computed as the difference of (log) prices P_t - P_{t-m} (or, again, using a return index in the place of prices). In this study, we focus on differences in log prices or index levels for simplicity.

Equivelance of Trend Following Systems

Of course, traders may want to define the TSMOM signal over a variety of horizons. In the extreme, one could use a series of daily (or monthly) returns and give each day's return a separate weight, which we call c:

$$TSMOM_t^c = \sum_{s=1}^{\infty} c_s \left(P_{t-s+1} - P_{t-s} \right) \tag{3}$$

This equation means that a general TSMOM signal can be generated by considering all the past daily price changes and assigning importance to each day based on how long ago it happened. For instance, one might want to rely more on recent price changes in assessing the current price trend. Ooi, Moskowitz, and Pedersen (2012) conduct a detailed analysis of how returns at various lags predict future returns. A trend-following strategy is characterized by having positive coefficients c_s, whereas negative coefficients correspond to reversal trades.

Equivelance of Trend Following Systems

Of course, this is just one example of an MACROSS strategy. Other strategies arise by varying the time horizons (here the 20 and 260 day averaging periods). Further, we need not give each day's price an equal weight in the moving average. Another common method is to weight past prices exponentially (called exponentially-weighted moving average as discussed in more detail later). More generally, we can compute the MA's using any weighting scheme, which we will denote by w (where the weights can for instance be equal weights or exponential weights as described in the examples further below). Hence, in general we can write the MA's mathematically as:

$$\begin{aligned} MA_t^{fast} &= \sum_{s=1}^{\infty} w_s^{fast} P_{t-s+1} \\ MA_t^{slow} &= \sum_{s=1}^{\infty} w_s^{slow} P_{t-s+1} \end{aligned} \tag{4}$$

The idea that one MA is faster than the other can be captured mathematically by the requirement that the fast MA places more weight on the most recent prices:

$$\sum_{j=1}^{s} w_j^{fast} \geq \sum_{j=1}^{s} w_j^{slow} \qquad \text{for all } s \tag{5}$$

Equivelance of Trend Following Systems

The trading signal is then the MACROSS, that is, the difference between these moving averages:

$$MACROSS_t = MA_t^{fast} - MA_t^{slow} \tag{6}$$

The MACROSS signal is the difference between two MA's and therefore a combination of past prices:

$$MACROSS_t = \sum_{s=1}^{\infty} \left(w_s^{fast} - w_s^{slow} \right) P_{t-s+1} \tag{7}$$

Equivelance of Trend Following Systems

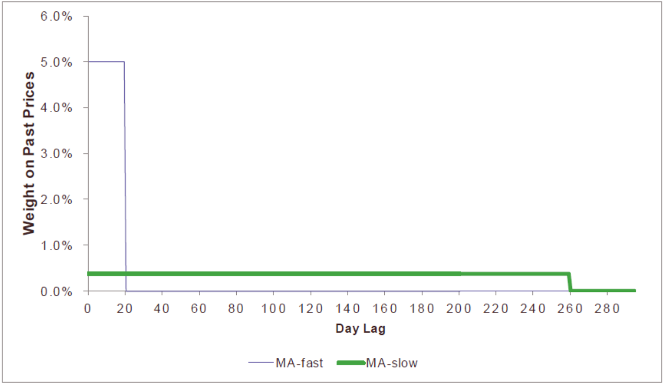
Specifically, the MACROSS equation (6) is equivalent to the TSMOM strategy (3) with coefficients on past returns c_s computed as follows[4]

$$c_s = \sum_{j=1}^{s} (w_j^{fast} - w_j^{slow}) \tag{8}$$

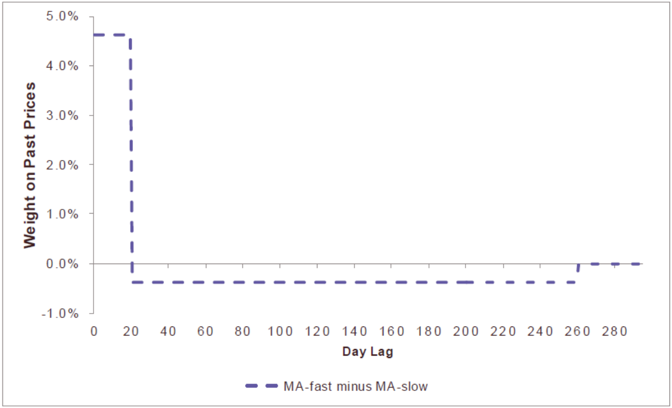
These implied coefficients c_s are positive for all MACROSS strategies where the fast MA is uniformly faster than the slow MA as given by Equation (5), which is true for the standard MACROSS strategies. It is natural that these TSMOM coefficients are positive since this means that the strategy is trend following (whereas negative coefficients would have indicated a bet on trend reversal).

Equivelance of Trend Following Systems

Figure 3. Moving-Average Crossover Coefficients: Fast and Slow Averages. *The fast moving average (MA) is an equal-weighted average over the past 20 days, while the slow MA is an equal-weighted average of the past 260 days.*

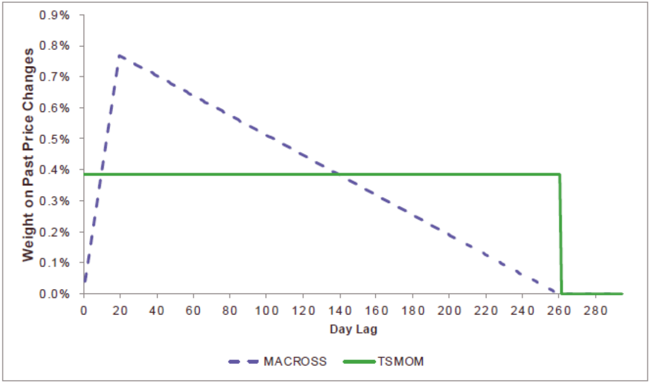


Equivelance of Trend Following Systems



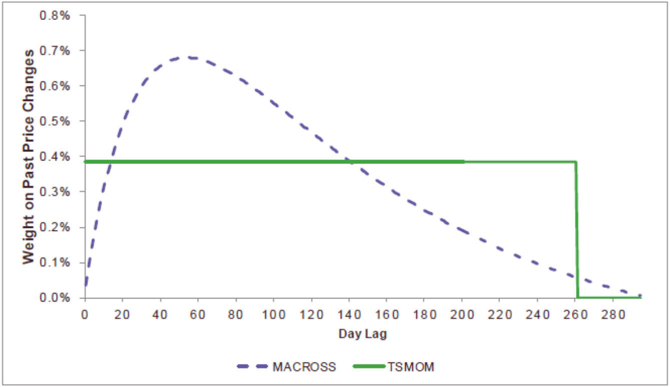
Equivelance of Trend Following Systems

Figure 5. Return Signature Plot: Trend Coefficients in Return Space. *This figure shows how much weight the trend indicator places on each daily return in the past for the simple 260-day time series momentum (TSMOM) and the 20/260 moving-average crossover (MACROSS), respectively, where the weights are normalized to sum to one. That is, a trend indicator can be viewed as an average of daily returns (rather than prices), as the figure shows.*



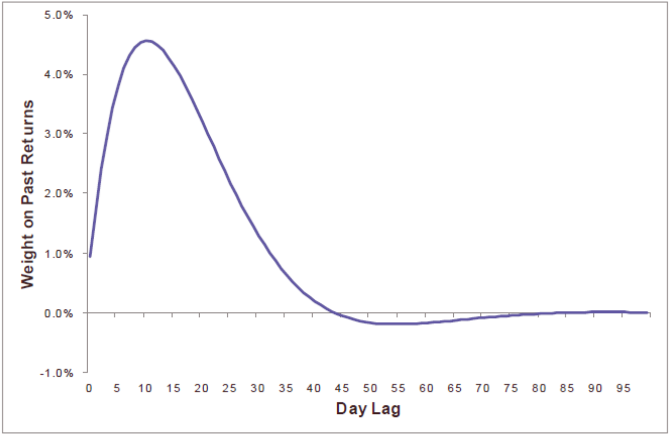
Equivelance of Trend Following Systems

Figure 7. Return Signature Plot: Moving-Average Crossover using Exponential Weights. *This figure shows the weights that an exponentially weighted moving average (EWMA) crossover puts on past returns. The centers of mass used for the EWMAs are 32 and 128 for the fast and slow EWMAs, respectively. For comparison, a 260 day time series momentum (TSMOM) signal is also shown. Weights have been normalized to sum to one in both cases.*



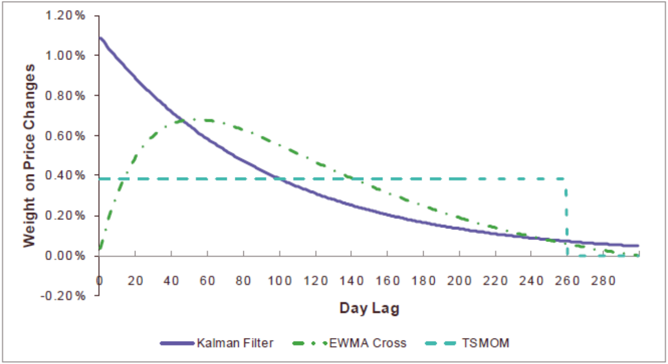
Equivelance of Trend Following Systems

Figure 9. Return Signature Plot: Hodrick and Prescott Filter. *This figure shows the weights on past returns for trend$_t$, for $\lambda=10^4$.*



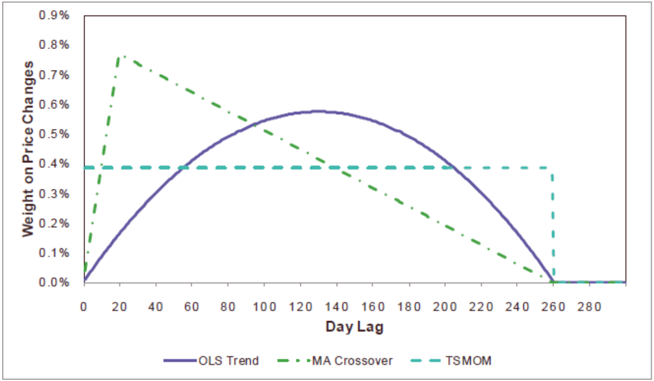
Equivelance of Trend Following Systems

Figure 10. Return Signature Plot: Kalman Filter. *This figure shows the weights on past price changes, or returns, for a Kalman Filter applied to a local linear trend model. In this case, a center of mass of 96 is used to form the exponential weights. Also shown are weights for an exponentially weighted moving average (EWMA) crossover with centers of mass of 32 and 128 for the fast and slow EWMAs, respectively, as well as weights for a simple 260 day time series momentum (TSMOM) signal.*



Equivelance of Trend Following Systems

Figure 13. Return Signature Plot: OLS Trend. *This figure shows the weights on past price changes, or returns, for the OLS trend estimator, compared to the simple 20/260 MACROSS signal and the simple 260-day TSMOM signal. Each of the three sets of weights is normalized to sum to one.*

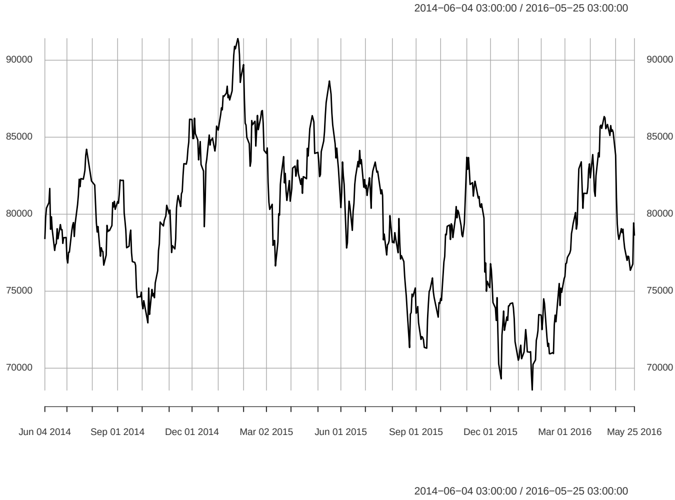


Adaptive Moving Averages

- Application of moving averages often involves a tradeoff between the amount of smoothness required and the amount of lag that can be tolerated.
- To obtain a better balance, various adaptive moving averages are proposed
- Exponential MA:
    - $EMA_t = \lambda P_t + (1-\lambda)EMA_{t-1}$
    - $\lambda = \frac{2}{1+n}$
    - λ is fixed, i.e. not adaptive
- Adaptive MAs
    - Make λ adaptive, i.e. $\lambda_t = f(P_{1:t})$
- Examples
    - Kaufman Adaptive Moving Average (KAMA)
    - FRAMA (Fractal Adaptive Moving Average)
    - Variable Moving Average (VMA)

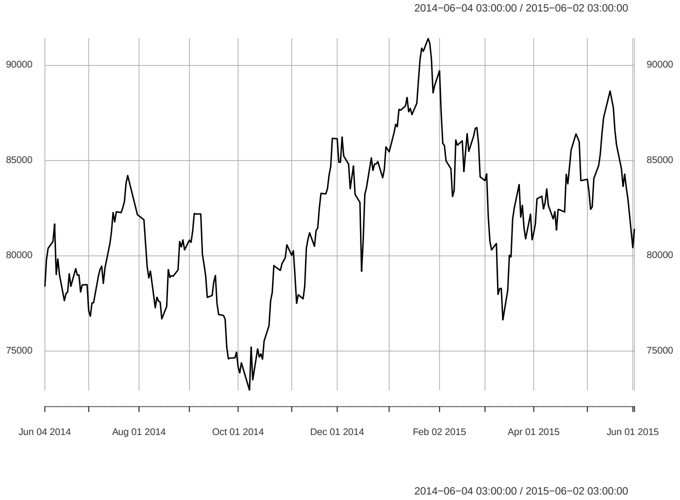
Adaptive Moving Averages

FRAMA (Fractal Adaptive Moving Average)



Adaptive Moving Averages

FRAMA (Fractal Adaptive Moving Average)



Advanced Trend Filtering Methods

Trend Filtering Methods for Momentum Strategies*

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December 2011

Abstract

This paper studies trend filtering methods. These methods are widely used in momentum strategies, which correspond to an investment style based only on the history of past prices. For example, the CTA strategy used by hedge funds is one of the best-known momentum strategies. In this paper, we review the different econometric estimators to extract a trend of a time series. We distinguish between linear and nonlinear models as well as univariate and multivariate filtering. For each approach, we provide a comprehensive presentation, an overview of its advantages and disadvantages and an application to the S&P 500 index. We also consider the calibration problem of

Advanced Trend Filtering Methods

be a noise or a stochastic cycle. Let y_t be a stochastic process. We assume that y_t is the sum of two different unobservable parts:

$$y_t = x_t + \varepsilon_t$$

where x_t represents the trend and ε_t is a stochastic (or noise) process. There is no precise definition for trend, but it is generally accepted to be a smooth function representing long-term movements:

> "*[...] the essential idea of trend is that it shall be smooth.*" (Kendall, 1973).

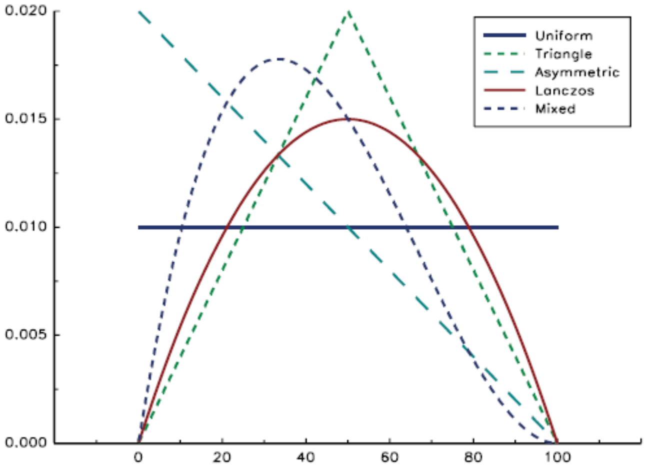
It means that changes in the trend x_t must be smaller than those of the process y_t. From a statistical standpoint, it implies that the volatility of $y_t - y_{t-1}$ is higher than the volatility of $x_t - x_{t-1}$:

$$\sigma\left(y_t - y_{t-1}\right) \gg \sigma\left(x_t - x_{t-1}\right)$$

One of the major problems in financial econometrics is the estimation of x_t. This is the subject of signal extraction and filtering (Pollock, 2009).

Advanced Trend Filtering Methods

Figure 2: Window function $\mathcal{L}_i$ of moving average filters ($n = 100$)



Advanced Trend Filtering Methods

If we consider a trend that is not constant, we may define the following objective function:

$$\frac{1}{2}\sum_{t=1}^{n}\left(y_t - \hat{x}_t\right)^2 + \lambda\sum_{t=2}^{n-1}\left(\hat{x}_{t-1} - 2\hat{x}_t + \hat{x}_{t+1}\right)^2$$

In this function, λ is the regularisation parameter which controls the competition between the smoothness[8] of $\hat{x}_t$ and the noise $y_t - \hat{x}_t$. We may rewrite the objective function in the vectorial form:

$$\frac{1}{2}\left\|y - \hat{x}\right\|_2^2 + \lambda\left\|D\hat{x}\right\|_2^2$$

where $y = (y_1, \ldots, y_n)$, $\hat{x} = (\hat{x}_1, \ldots, \hat{x}_n)$ and the D operator is the $(n-2) \times n$ matrix:

$$D = \begin{bmatrix} 1 & -2 & 1 & & & & \\ & 1 & -2 & 1 & & & \\ & & & \ddots & & & \\ & & & 1 & -2 & 1 & \\ & & & & 1 & 2 & 1 \end{bmatrix}$$

The estimator is then given by the following solution:

$$\hat{x} = \left(I + 2\lambda D^{\top} D\right)^{-1} y$$

It is known as the Hodrick-Prescott filter (or L_2 filter). This filter plays an important role in calibrating the business cycle.

Advanced Trend Filtering Methods

Kalman filtering Another important trend estimation technique is the Kalman filter, which is described in Appendix A.1. In this case, the trend μ_t is a hidden process which follows a given dynamic. For example, we may assume that the model is[9]:

$$\begin{cases} R_t = \mu_t + \sigma_\zeta \zeta_t \\ \mu_t = \mu_{t-1} + \sigma_\eta \eta_t \end{cases} \tag{6}$$

Here, the equation of R_t is the measurement equation and R_t is the observable signal of realised returns. The hidden process μ_t is supposed to follow a random walk. We define $\hat{\mu}_{t|t-1} = \mathbb{E}_{t-1}[\mu_t]$ and $P_{t|t-1} = \mathbb{E}_{t-1}\left[\left(\hat{\mu}_{t|t-1} - \mu_t\right)^2\right]$. Using the results given in Appendix A.1, we have:

$$\hat{\mu}_{t+1|t} = (1 - K_t)\,\hat{\mu}_{t|t-1} + K_t R_t$$

where $K_t = P_{t|t-1} / \left(P_{t|t-1} + \sigma_\zeta^2\right)$ is the Kalman gain. The estimation error is determined by Riccati's equation:

$$P_{t+1|t} = P_{t|t-1} + \sigma_\eta^2 - P_{t|t-1} K_t$$

Riccati's equation gives us the stationary solution:

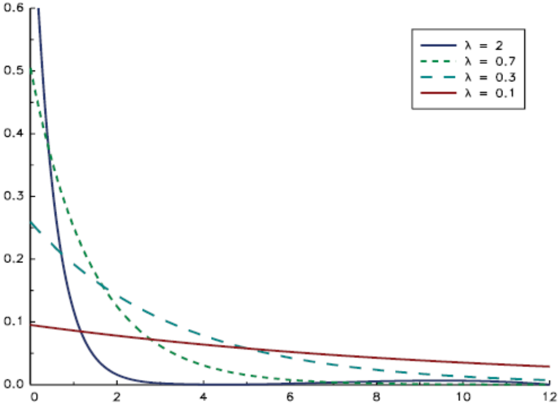
$$P^* = \frac{\sigma_\eta}{2}\left(\sigma_\eta + \sqrt{\sigma_\eta^2 + 4\sigma_\zeta^2}\right)$$

The filter equation becomes:

$$\hat{\mu}_{t+1|t} = (1 - \kappa)\,\hat{\mu}_{t|t-1} + \kappa R_t$$

Advanced Trend Filtering Methods

Figure 5: Window function $\mathcal{L}_i$ of the Kalman filter



[10]We have $0 < \kappa < 1$ and $lambda > 0$.
[11]We notice that $\hat{\mu}_{t+1|t} = \hat{\mu}_t$.

Advanced Trend Filtering Methods

We may wonder what the link is between the regression model (5) and the Markov model (6). Equation (5) is equivalent to the following state space model[12]:

$$\begin{cases} y_t = x_t + \sigma_\varepsilon \varepsilon_t \\ x_t = x_{t-1} + \mu \end{cases}$$

If we now consider that the trend is stochastic, the model becomes:

$$\begin{cases} y_t = x_t + \sigma_\varepsilon \varepsilon_t \\ x_t = x_{t-1} + \mu + \sigma_\zeta \zeta_t \end{cases}$$

This model is called the local level model. We may also assume that the slope of the trend is stochastic, in which case we obtain the local linear trend model:

$$\begin{cases} y_t = x_t + \sigma_\varepsilon \varepsilon_t \\ x_t = x_{t-1} + \mu_{t-1} + \sigma_\zeta \zeta_t \\ \mu_t = \mu_{t-1} + \sigma_\eta \eta_t \end{cases}$$

Advanced Trend Filtering Methods

In the regression model (5), we assume that $x_t = f(t)$ while $f(t) = \mu t$. The model is said to be parametric because the estimation of the trend consists of estimating the parameter μ. We then have $\hat{x}_t = \hat{\mu} t$. With nonparametric regression, we directly estimate the function f, obtaining $\hat{x}_t = \hat{f}(t)$. Some examples of nonparametric regression are kernel regression, loess regression and spline regression. A popular method for trend filtering is local polynomial regression:

$$\begin{aligned} y_t &= f(t) + \varepsilon_t \\ &= \beta_0(\tau) + \sum_{j=1}^{p} \beta_j(\tau)(\tau - t)^j + \varepsilon_t \end{aligned}$$

For a given value of τ, we estimate the parameters $\hat{\beta}_j(\tau)$ using weighted least squares with the following weightings:

$$w_t = \mathcal{K}\left(\frac{\tau - t}{h}\right)$$

where $\mathcal{K}$ is the kernel function with a bandwidth h. We deduce that:

$$\hat{x}_t = \mathbb{E}[y_t | \tau = t] = \hat{\beta}_0(t)$$

Advanced Trend Filtering Methods

discussed in the work of Daubechies *et al.* (2004) in relation to the linear inverse problem, while Tibshirani (1996) considers the Lasso regression problem. If we consider an L_1 filter, the objective function becomes:

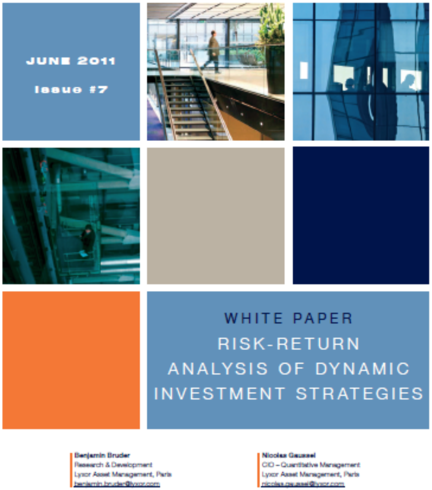
$$\frac{1}{2}\sum_{t=1}^{n}(y_t - \hat{x}_t)^2 + \lambda\sum_{t=2}^{n-1}|\hat{x}_{t-1} - 2\hat{x}_t + \hat{x}_{t+1}|$$

which is equivalent to the following vectorial form:

$$\frac{1}{2}\|y - \hat{x}\|_2^2 + \lambda\|D\hat{x}\|_1$$

Kim *et al.* (2009) shows that the dual problem of this L_1 filter scheme is a quadratic programme with some boundary constraints[20]. To find $\hat{x}$, we may also use the quadratic programming algorithm, but Kim *et al.* (2009) suggest using the primal-dual interior point method in order to optimise the numerical computation speed.

Theoretical Return Profile for Trend Following Systems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Issue #7

WHITE PAPER

RISK-RETURN ANALYSIS OF DYNAMIC INVESTMENT STRATEGIES

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Theoretical Return Profile for Trend Following Systems

2.1 Model and results

Consider a simplified situation in which an asset can be traded at each date t at price S_t, without any friction of any kind. S_t is supposed to be governed by an ordinary diffusion model:

$$\begin{aligned} \mathrm{d}S_t &= S_t \times (\mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t) \\ S_0 &= s \end{aligned}$$

An investor is running an investment strategy consisting in holding a number $f(S_t)$ of securities at any time. This strategy is supposed to depend only on the price itself and some parameters, but not on time or on other state variables. For the sake of simplicity, it is assumed that interest rates are zero[1]. The wealth of the investor at each date t is denoted X_t. On a day-to-day basis, or between t and $t + \mathrm{d}t$, variation in wealth is written as:

$$\mathrm{d}X_t = f(S_t)\,\mathrm{d}S_t \tag{1}$$

If S_t was deterministic and non-stochastic, it would be clear that:

$$\begin{aligned} X_T - X_0 &= \int_{S_0}^{S_T} f(S_t)\,\mathrm{d}S_t \\ &= F(S_T) - F(S_0) \end{aligned}$$

where:

$$F(S) \equiv \int_a^S f(x)\,\mathrm{d}x,$$

whatever the a chosen. If it is not, Ito's lemma applied to the function F yields the important property we want to emphasise

Theoretical Return Profile for Trend Following Systems

Proposition 1 *Any portfolio strategy of the type (1) described above can be broken down into an option profile plus some trading impact as follows:*

$$X_T - X_0 = \underbrace{F(S_T) - F(S_0)}_{option\ profile} + \underbrace{-\frac{1}{2}\int_0^T f'(S_t)\, S_t^2 \sigma_t^2\, \mathrm{d}t}_{trading\ impact} \tag{2}$$

While simple, the above proposition yields some interesting qualitative properties. Regarding model assumptions, it is worth noticing the robustness of this property, which can be obtained whenever Ito's lemma can be used with continuous price processes. This covers a wide variety of models and does not rely on special probabilistic assumptions. These include Black-Scholes, of course, but also local and stochastic volatility models. On the other hand, the option profile obtained is European. If the strategy is richer in terms of variable states, the option profile will be a function of those different states.

Theoretical Return Profile for Trend Following Systems

Our purpose here is not to find the best estimate, but to provide clear analysis. We will therefore consider the simple example of a moving average of past daily returns. We choose a moving average with exponential weights:

$$\hat{\mu}_t = \frac{1}{\tau} \sum_{i=0}^{n} e^{-\frac{i\delta t}{\tau}} \frac{S_{t-i\delta t} - S_{t-(i+1)\delta t}}{S_{t-(i+1)\delta t}}$$

Such an estimator has interesting properties. It depends on a single parameter τ, which represents the average duration of estimation. Due to exponential weighting, recent returns have a larger impact than older ones. It does not suffer from "threshold effect", that is, changes of regimes due to a past observation that exits the averaging window. Moreover, this estimator has theoretical foundations, as it can be interpreted as the Kalman filter for an

Theoretical Return Profile for Trend Following Systems

In the following, we suppose that the investor considers that the best returns estimation between times t and $t + \mathrm{dt}$ is $\hat{\mu}_t \, \mathrm{dt}$. Based on this assumption, the investor can simply apply the optimal Markowitz/Merton strategy[9]. Here, we will still consider that the risk-free rate is 0. In this case, its exposure to the risky asset will be:

$$e_t = m \frac{\hat{\mu}_t}{\sigma^2} \tag{9}$$

where μ is the trend estimator, σ is the annualised volatility of the underlying, and m is a risk tolerance parameter. Interestingly, this exposure strategy is qualitatively consistent with the principles described in the Turtles Rules. The exposure to the risky asset is proportional to this risk tolerance. This parameter depends on the investor's risk profile. This exposure is also proportional to the trend estimator value $\hat{\mu}_t$ and inversely proportional to the risk. Note that it is not capped and is almost never 0.

Theoretical Return Profile for Trend Following Systems

Following this strategy, the investor's wealth evolves as equation (3):

$$\begin{aligned} \frac{\mathrm{d}X_t}{X_t} &= e_t \frac{\mathrm{d}S_t}{S_t} \\ &= m \frac{\hat{\mu}_t}{\sigma^2} \frac{\mathrm{d}S_t}{S_t} \end{aligned} \tag{10}$$

Now, by inverting equation (8), the dynamic of the wealth can be written as a function of the trend $\hat{\mu}_t$. We can then follow the steps of Proposition 2 where the variable is no longer the asset price S_t but the asset trend $\hat{\mu}_t$.

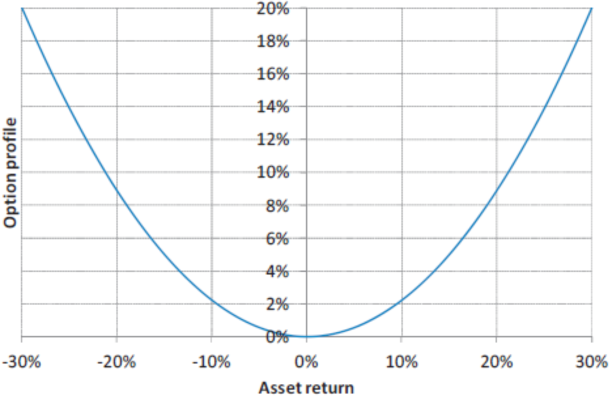
Proposition 4 *The logarithmic return of a trend-following strategy that follows the exposure function (9) can be broken down into an option profile and some trading impact:*

$$\ln \frac{X_T}{X_0} = m \left(\underbrace{\frac{\tau}{2\sigma^2} \left(\hat{\mu}_T^2 - \hat{\mu}_0^2 \right)}_{\textit{option profile}} + \underbrace{\int_0^T \left[\frac{\hat{\mu}_t^2}{\sigma^2} \left(1 - \frac{1}{2} m \right) - \frac{1}{2\tau} \right] \mathrm{d}t}_{\textit{trading impact}} \right) \tag{11}$$

It is worth remembering that the trend $\hat{\mu}_t$ is not a model assumption but the actual measured trend.

Theoretical Return Profile for Trend Following Systems

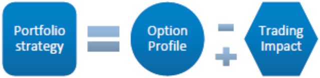
Figure 17: Short term option profile, as a function of the risky asset return



Theoretical Return Profile for Trend Following Systems

Qualitatively speaking, the option hedging paradigm can be represented as follows:

whereas the fund management situation can be represented as follows:



Technically speaking, the option profile is the integral of the holding function while the trading impact is related to the derivative of the holding function. These observations are summed up in the following table:

Strategy Type	Option Profile	Trading Impact	Hit Ratio	Average Gain / Loss Comparison
Buying when market goes up	Convex	Negative	$\lesssim 50\%$	Average Gain > Average Loss
Buying when market goes down	Concave	Positive	$\gtrsim 50\%$	Average Loss > Average Gain

Impact of Volatility & Correlation Regimes

When Do Trend Followers Make Money?

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Portfolio Manager
Ellington Quantitative Strategies

March 2013

Impact of Volatility & Correlation Regimes

	Vol Low	Vol High
Cor Low	0.9	0.1
Cor High	0.0	1.1

Table 1: Barclay CTA Index Sharpe Ratio by Vol-Cor State. Data from Jan 1991 - Jan 2013.

Impact of Volatility & Correlation Regimes

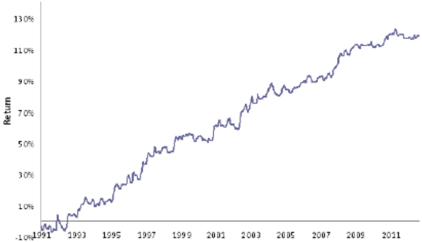


Figure 4: Barclay CTA Index performance

Impact of Volatility & Correlation Regimes

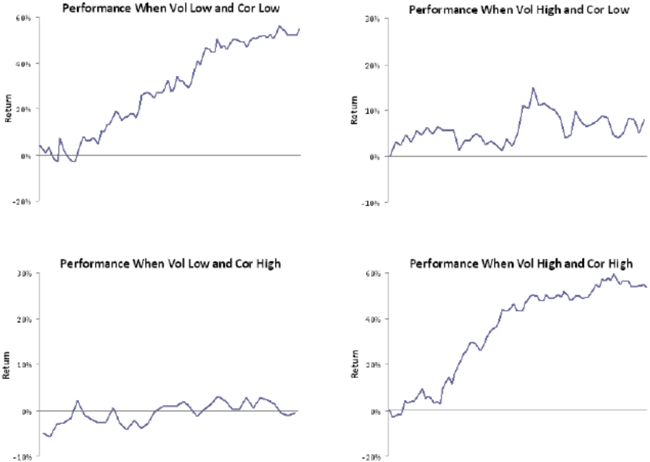


Figure 5: Barclay CTA Index Performance by Vol-Cor State

Impact of Volatility & Correlation Regimes

Low Volatility, Low Correlation. In low Volatility environments, assets do not react violently to information arrival. This increases the likelihood of *underreaction* to news (and thereby follow-on momentum), which is good for trend followers. In addition, low Correlation implies that related markets are not moving in lock-step. For instance, suppose that a crop report sends corn but not lean hogs higher. Eventually, high corn prices will cause higher hog prices. Thus, low Correlation makes it more likely that information diffuses across markets slowly. This gives trend followers an opportunity to exploit the coming trend in lean hogs (implicitly capitalizing on lead-lag relationships). We expect this environment to be ideal for trend following, and Figure 5 bears this out.[9]

High Volatility, Low Correlation. As argued above, low Correlation implies that trend followers can profit from lead-lag relationships. But the above also implies that high Volatility increases the likelihood of *overreaction* to news (and hence reversals), which is bad for trend followers. These two forces compete, leaving trend followers treading water in this state (Figure 5).

Impact of Volatility & Correlation Regimes

Low Volatility, High Correlation. High Correlation reduces lead-lag profits. It also causes trend followers to accumulate large net exposures to asset classes at the worst times. To see this, suppose positive economic data drives US equities persistently higher. Since correlations are high within asset classes, all equity indices move up together. Soon, trend followers are very long global equities. The problem is that, because *across* asset class Correlation is high, any news that moves commodities or bonds would also cause equities to move in lock-step. Thus, trend followers become most exposed precisely when likelihood of exogenous shocks is highest. Low Volatility, as we saw, provides asset-specific momentum opportunities, making this state overall a wash for trend followers.

High Volatility, High Correlation. This state indicates crisis. A good example is 2008, and before that the state materialized in (e.g.) 1998, the 2001-2002 recession, and during the pullback in 2004. High Volatility reduces asset-specific drift. High Correlation diminishes lead-lag profits and induces large net exposures to asset classes. These forces are all directionally bad for trend followers. But the big opposing force is that crisis causes persistent deleveraging, driving positively autocorrelated asset class returns. In this state, the large net exposures can actually pay off. So long as deleveraging happens over an extended window, CTAs will benefit as they tactically get long or short in the middle of a cycle of outflows. We can see from Figure 5 that CTAs did well in this state in the early part of the sample but have been struggling towards the end of the sample, suggesting that deleveraging speeds have been increasing since 2008.

Impact of Volatility & Correlation Regimes

Table 5: Performance (t-statistics*) of Cross Asset Prototype Absolute Momentum Factors under different macro/market regimes

	Growth			Inflation			Volatility			Liquidity		
	Low	**Mid**	**High**	**Low**	**Mid**	**High**	**Low**	**Mid**	**High**	**Low**	**Mid**	**High**
Equity - 12 Months	2.87	15.97	14.10	5.32	17.48	9.05	19.14	11.53	2.21	8.81	6.99	17.03
	(-2.04)	(1.25)	(0.79)	(-1.34)	(1.75)	(-0.47)	(2.06)	(0.14)	(-2.21)	(-0.54)	(-0.99)	(1.53)
Bond - 12 Months	17.24	12.98	-0.25	12.41	6.05	12.05	9.71	-2.73	22.81	14.88	0.61	14.33
	(1.70)	(0.69)	(-2.41)	(0.54)	(-0.98)	(0.47)	(-0.06)	(-2.98)	(3.04)	(1.14)	(-2.18)	(1.02)
Curncy - 12 Months	-0.20	10.47	7.66	2.02	10.82	4.29	7.99	6.60	3.30	4.07	3.00	10.77
	(-2.06)	(1.49)	(0.57)	(-1.25)	(1.73)	(-0.54)	(0.67)	(0.21)	(-0.89)	(-0.63)	(-0.97)	(1.61)
Comdty - 12 Months	3.21	9.51	7.65	5.18	9.04	5.76	11.07	5.90	3.34	7.08	4.11	9.08
	(-1.14)	(0.87)	(0.28)	(-0.48)	(0.77)	(-0.32)	(1.38)	(-0.28)	(-1.10)	(0.10)	(-0.84)	(0.74)

Impact of Volatility & Correlation Regimes

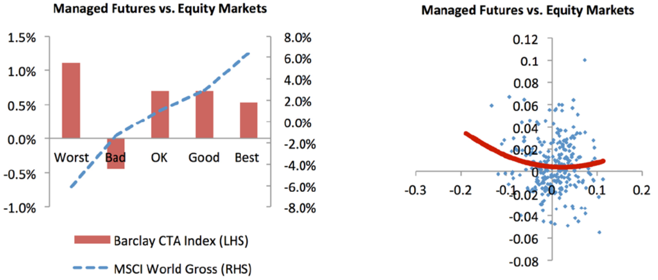
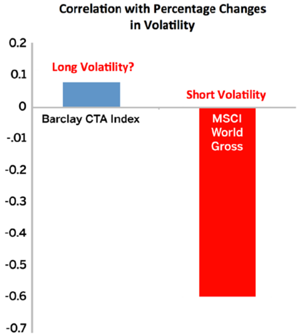


Figure 4: Managed Futures (Barclay CTA Index) vs. Equity Markets (MSCI World Gross) Bar Chart and Scatterplot. Source: Pertrac and HFR

Impact of Volatility & Correlation Regimes



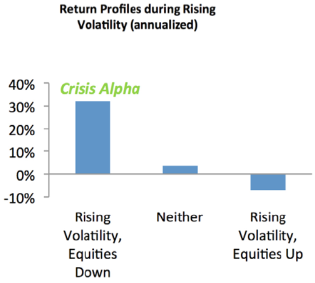


Figure 5: Correlations with Changes in Volatility: Managed Futures (Barclay CTA Index) and MSCI World Gross. Source: Pertrac and HFR

Figure 6: Return Profiles during Rising Volatility for Managed Futures Source: Pertrac and HFR

Trend Factor for Stock Selection

A Trend Factor: Any Economic Gains from Using Information over Investment Horizons?

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Current version: September, 2015

(Han, Zhou, and Zhu 2015)

Trend Factor for Stock Selection

Abstract

In this paper, we provide a trend factor that captures simultaneously all three stock price trends: the short-, intermediate- and long-term. It outperforms substantially the well-known short-term reversal, momentum and long-term reversal factors, which are based on the three price trends separately, by more than doubling their Sharpe ratios. During the recent financial crisis, the trend factor earns 0.75% per month, while the market loses −2.03% per month, the short-term reversal factor loses −0.82%, the momentum factor loses −3.88% and the long-term reversal factor barely gains 0.03%. The performance of the trend factor is robust to alternative formations and to a variety of control variables. From an asset pricing perspective, it performs well too in explaining cross-section stock returns.

Trend Factor for Stock Selection

2.2. *Methodology*

To construct the trend factor, we first calculate the MA prices on the last trading day of each month. The MA on the last trading day of month t of lag L is defined as

$$A_{jt,L} = \frac{P^t_{j,d-L+1} + P^t_{j,d-L+2} + \cdots + P^t_{j,d-1} + P^t_{j,d}}{L}, \tag{1}$$

where P^t_{jd} is the closing price for stock j on the last trading day d of month t, and L is the lag length. Then, we normalize the moving average prices by the closing price on the last

5

trading day of the month,

$$\tilde{A}_{jt,L} = \frac{A_{jt,L}}{P^t_{jd}}. \tag{2}$$

Trend Factor for Stock Selection

To predict the monthly expected stock returns cross-sectionally, we use a two-step procedure. In the first-step, we run in each month t a cross-section regression of stock returns on observed normalized MA signals to obtain the time-series of the coefficients on the signals,

$$r_{j,t} = \beta_{0,t} + \sum_i \beta_{i,t} \tilde{A}_{jt-1,L_i} + \epsilon_{j,t}, \quad j = 1, \cdots, n, \tag{3}$$

where

$$\begin{aligned} r_{j,t} &= \text{rate of return on stock } j \text{ in month } t, \\ \tilde{A}_{jt-1,L_i} &= \text{trend signal at the end of month } t-1 \text{ on stock } j \text{ with lag } L_i, \\ \beta_{i,t} &= \text{coefficient of the trend signal with lag } L_i \text{ in month } t, \\ \beta_{0,t} &= \text{intercept in month } t, \end{aligned}$$

and n is the number of stocks.[3] It should be mentioned that only information in month t or prior is used above to regress returns in month t.

Trend Factor for Stock Selection

Following, for example, Brock, Lakonishok, and LeBaron (1992), we consider in the above regressions using MAs of lag lengths 3-, 5-, 10-, 20-, 50-, 100-, and 200-days. In addition, we include 400-, 600-, 800- and 1000-days. These MA signals indicate the daily, weekly, monthly, quarterly, 1-year, 2-year, 3-year, and 4-year price trends of the underlying stock. Note that we use simply all the popular MA indicators without any alterations or optimization to mitigate concerns of data-mining, an issue to be examine further in Section 3.

Trend Factor for Stock Selection

Then, in the second-step, as in Haugen and Baker (1996), we estimate the expected return for month $t+1$ from

$$E_t[r_{j,t+1}] = \sum_i E_t[\beta_{i,t+1}]\tilde{A}_{jt,L_i}, \tag{4}$$

Trend Factor for Stock Selection

where $E_t[r_{j,t+1}]$ is our forecasted expected return on stock j for month $t+1$, and $E_t[\beta_{i,t+1}]$ is the estimated expected coefficient of the trend signal with lag L_i, and is give by

$$E_t[\beta_{i,t+1}] = \frac{1}{12}\sum_{m=1}^{12}\beta_{i,t+1-m}, \tag{5}$$

which is the average of the estimated loadings on the trend signals over the past 12 months.

Trend Factor for Stock Selection

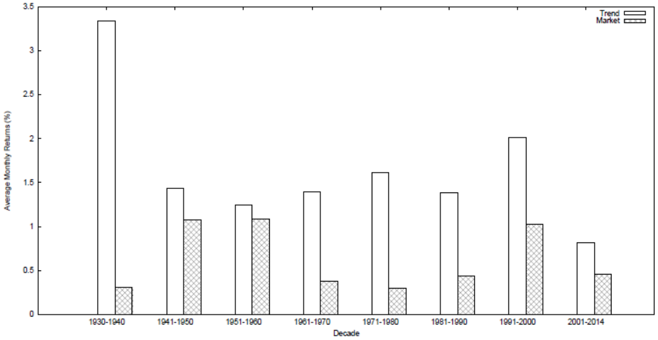


Figure 1: Trend Factor Performance in Subperiods
This figure plots the average monthly returns of the trend factor and the market over roughly each of the past eight decades. The first period is from June 1930 to December 1940, the second is from January 1941 to December 1950, and the last is from January 2000 to December 2014.

Trend Factor for Stock Selection

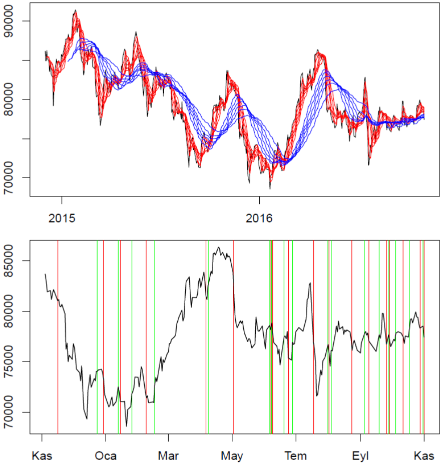
Table 1: The Trend Factor and Other Factors: Summary Statistics

This table reports the summary statistics of the trend factor (*Trend*), short-term reversal factor (*SREV*), momentum factor (*MOM*), long-term reversal factor (*LREV*), and Fama-French three factors including the market portfolio (*Market*), *SMB* and *HML* factors. We report the summary statistics such as sample mean in percentage, sample standard deviation in percentage, Sharpe ratio, skewness, and excess kurtosis. Significance at the 1%, 5%, and 10% levels is given by an ***, an **, and an *, respectively. The sample period is from June, 1930 through December, 2014.

Factor	Mean(%)	Std Dev(%)	Sharpe Ratio	Skewness	Excess Kurtosis
Trend	1.63	3.45	0.47	1.47	11.3
SREV	0.79	3.49	0.23	0.99	8.22
MOM	0.79	7.69	0.10	-4.43	40.7
LREV	0.34	3.50	0.10	2.93	24.8
Market	0.62	5.40	0.12	0.27	8.03
SMB	0.27	3.24	0.08	2.04	19.9
HML	0.41	3.58	0.11	2.15	18.9

Money Management

Typical Return Profile



Money Management

Position Sizing

- **Pyramiding** refers to the position sizing strategy of entering a fractional amount of your intended position size first and as price moves in your favor, you add to the winning position; and ideally move your stop loss to protect your profits so far.
  - On a fake breakout, your position will be relatively small because you haven't yet reached the full position.
  - Only when the breakout is strong and successful you reach your maximum position size and fully capitalize on winning trades.
- Scaling in vs Scaling out

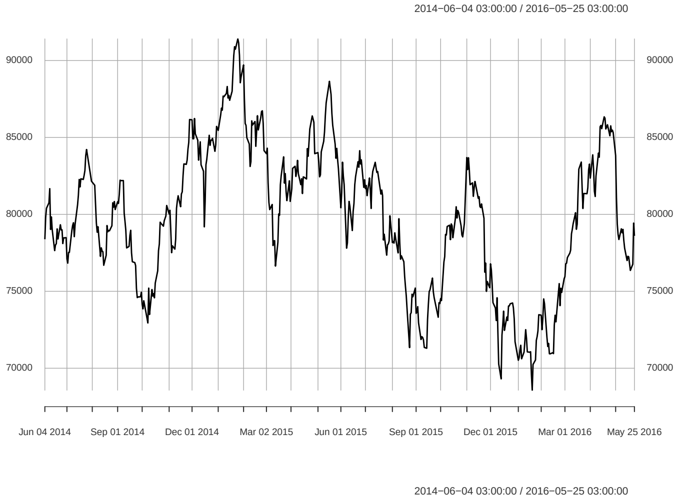
Money Management

Countertrend Entry

- When a stock price breaks a resistance level, old resistance becomes new support. When a stock breaks a support level, old support becomes new resistance.
- In the majority of your trades, the stock will test the level it has broken after the first couple of days.
- Thus countertrend moves within a trend can be better entry points

Money Management

Countertrend Entry



Assignment

Backtesting with `quantstrat`

- **Trade Idea**: We will test two trading strategies based on return runs.
  - Trend Following Strategy:
    - Buy after n days of positive return
    - Sell after m days of negative return
  - Mean Reversion Strategy:
    - Buy after n days of negative return
    - Sell after m days of positive return
- Design two different trading algo for these strategies in `quantstrat`
- Apply these strategies to BIST100 index data
- For each strategy, optimize n and m using a grid search over set $1, 2, ..., 10$
- Compare optimized versions of the two strategies
- Is it better to be a trend-follower or a contrarian in BIST?

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